In application, differential equations do not arise by eliminating the arbitrary constants. They frequently arise while investigating many physical problems in all fields of engineering, science and even in social sciences. Mostly these differential equations are also accompanied by certain conditions on the variables to obtain unique solution satisfying the given conditions.

Example 10.7

Show that $x^2 + y^2 = r^2$, where r is a constant, is a solution of the differential equation $\frac{dy}{dx} = -\frac{x}{y}$.

Solution

Given that $x^2 + y^2 = r^2, r \in \mathbb{R}$ ... (1)

The given equation contains exactly one arbitrary constant.

So, we have to differentiate the given equation once. Differentiate (1) with respect to x, we get

$$2x + 2y\frac{dy}{dx} = 0, \text{ which implies } \frac{dy}{dx} = -\frac{x}{y}.$$

Thus, $x^2 + y^2 = r^2$ satisfies the differential equation $\frac{dy}{dx} = -\frac{x}{y}$.

Hence, $x^2 + y^2 = r^2$ is a solution of the differential equation $\frac{dy}{dx} = -\frac{x}{y}$.

Example 10.8

Show that $y = mx + \frac{7}{m}, m \neq 0$ is a solution of the differential equation $xy' + 7\frac{1}{y'} - y = 0$.

Solution

The given function is $y = mx + \frac{7}{m}$, where m is an arbitrary constant. ... (1)

Differentiating both sides of equation (1) with respect to x, we get $y' = m$.

Substituting the values of y' and y in the given differential equation,

we get $xy' + \frac{7}{y'} - y = xm + \frac{7}{m} - mx - \frac{7}{m} = 0$.

Therefore, the given function is a solution of the differential equation $xy' + 7\frac{1}{y'} - y = 0$.

Example 10.9

Show that $y = 2\left(x^2 - 1\right) + Ce^{-x^2}$ is a solution of the differential equation $\frac{dy}{dx} + 2xy - 4x^3 = 0$.

Solution

The given function is $y = 2\left(x^2 - 1\right) + Ce^{-x^2}$, where C is an arbitrary constant. ...(1)

Differentiating both sides of equation (1) with respect to x, we get $\frac{dy}{dx} = 4x - 2xCe^{-x^2}$.

Substituting the values of $\frac{dy}{dx}$ and y in the given differential equation, we get

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